13/8/26. EXPERIMENT 6.

Error detection & Error Correction.

AIM:

Write a program to implement error detection and correction using hamming code concept. Make a test run to input data stream

Sender program:

* Input to sender file should be a text of any length.
* Apply hamming code concept on the binary data and add redudant bits to it.
* Save in a file called channel.

Code (Sender.py)

```
data = list (map (int, input ("Enter data bits :")
m = len (data)
r = 0
while (2**r) < (m+r+1)
    r += 1
n = m+r
hamming = [0]*(n+1)
j = 0
for (i in range (1, n+1):
```

```
if (i (j-1)) != 0;
    hamming [i] = data [j]
        j+=1
for p in range (r):
    pos = 2 ** p
    parity = 0

print ("Data bits : data)
print ("Hamming code : code)
```

Output:

Enter data : 1011

Data bits : [1, 0, 1, 1]

Hamming code : [0, 1, 1, 0, 0, 1, 1].

Receiver code (receiver.py)

```
received = list (map (input ("Enter received
                hamming code ") strip ()))

n = len (received)
r = 0
while (2 ** r) i = n
    r+ = 1
error pos = 0
for p in range (n):
pos = 2 ** p
parity = 0
for i in range (1, n+1):
        if i pos:
            parity a = received (i-1)
        print ("No error detected")
```

Output:

Suppose hc. 0110011 change to 0111011

Error detected at position : 4

Corrected hamming code : [0, 1, 1, 1, 0, 1, 1]

Original data bits : [1, 0, 1, 1].

RESULT: Thus the hamming code algorithm has successfully implemented in python. The sender generated the hamming code from the given data bits.